FINANCIAL TECHNOLOGY · QUALITY ENGINEERING

Test Process: From Story to Release

Requirement Analysis & Test Design → Sprint Development → Sprint SIT → UAT → UAT Final Regression → Pre-release Regression → LV

Phase 1 — Requirement Analysis & Test Design

Owner: QA | Reviewer: PO team

QA

Analyze Requirements & Business Rules

Write SIT Test Cases

Tag the Acceptance Tests (AT) for PO Team Review

Select Automation Candidates

PO Team

Clarify Requirements & Answer Open Questions

Review Business Coverage

Approve the Acceptance Tests (AT)

✓ **Test Design Exit Gate**

PO team approved the AT. SIT test cases are ready.

Phase 2 — Sprint Development

Owner: FE / BE Developers / QA

FE Developer

Unit Tests

Integration Tests (Mocked UI E2E, Component)

Smoke Tests from QA Test Cases (manual, mocked back end)

BE Developer

Unit Tests

Integration Tests (API, Database, Downstream Services)

API Smoke Tests from QA Test Cases

QA

Prepare Test Data

Prepare Automation (Page Objects, Step Definitions)

✓ **SIT Entry Gate**

Developer smoke tests passed; environment and test data ready.

Phase 3 — Sprint SIT

Owner: QA

✓ Sanity Test

Execute key SIT cases on each new build to validate core features and real cross-system integration

✓ Exploratory Testing

Additional negative, corner and exception scenarios

✓ SIT Automation

Complete Sprint automation and integrate into the SIT Regression Suite

Defect Fix & Verification

Developer fixes → QA retests → QA runs impacted SIT regression tests.

✓ SIT Exit Gate

Functional, integration and SIT regression tests passed; residual risks accepted.

Phase 4 — UAT

Owner: PO Team / QA

PO Team

Execute Business Acceptance Tests

Defect Verification & Retesting

Business Test Evidence

QA

Consolidate PO Team-defined User Journeys into the UAT Regression Suite

Reuse & Compose Automation Scripts

Debug & Validate in UAT

Defect Fix & Verification

Developer fixes → PO team retests → QA runs impacted regression tests.

Phase 5 — UAT Final Regression

Owner: QA / PO Team

- ✓ Execute Automated User Journey Regression Suite
- ✓ Verify Defect Resolutions & Regression Results
- ✓ Generate Regression Reports & Test Evidence

✓ **UAT Exit Gate**

Regression passed, critical defects resolved or accepted, PO team sign-off obtained.

↓

Phase 6 — Pre-release Regression & Release Readiness

Owner: QA / Developers / Release Team

- ✓ **Pre-release Regression**
  QA runs the UAT regression suite to protect the existing features
- ✓ **Performance Test & Penetration Test**
  Developers run them once per release, not in each sprint
- ✓ **Release Readiness Check**
  Release team checks the deployment plan, the rollback plan, and the open defects

✓ **Go-live Gate**

Regression, performance test, and penetration test passed; release approved.

↓

After Release — LV (Live Verification)

Owner: Business Team · Outside the QA scope

The business team verifies the release in production.

Automation Suites in Ongoing Use

Owner: QA

SIT Regression Suite

Runs daily in SIT via CI/CD

Monitors the SIT environment and finds broken features early

Grows each sprint with the new SIT automation

UAT Regression Suite

Runs before each new release

Protects the existing features from the new changes

Grows each release with the new user journeys from UAT

Automation Maintenance

QA repairs failed tests, removes tests for retired features, and keeps the suites reliable.

Ownership Summary

QA (Test Design)

SIT Test Cases, AT Tagging & Automation Candidates

Developers

Unit & Integration Tests, Smoke Tests, Performance & Penetration Tests

QA (SIT)

System Validation & SIT Automation

PO Team

Requirement Clarification, AT Approval & Business Acceptance

QA (Regression)

User Journey Automation, Pre-release Regression & Suite Maintenance

Release Team

Release Readiness & Deployment

Business Team

LV in Production

Note: The acceptance tests (AT) are a tagged subset of the SIT test cases, not a separate document. QA tags them, the PO team approves them, and the PO team runs them in UAT. SIT and UAT have separate regression gates. Performance and penetration tests run once per release, not in each sprint. Defect retesting and impacted regression take place throughout the delivery lifecycle. SIT can progress on a sprint cadence while UAT follows the release cadence, provided builds and environments are controlled.